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Fast-Track Rigless Gas Lift Retrofit Using Non-Hazardous Kinley Perforator Technology and Advanced Production Analytics: A Case Study from Offshore Oman

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Abstract

This paper presents a detailed case study of a fast-track, rigless gas lift retrofit using the non-hazardous Perforator system—a non-hazardous, globally deployable solution—integrated with advanced production analytics to optimize gas lift performance in offshore oil wells. The retrofit was implemented in response to Electrical Submersible Pump (ESP) failures in two offshore Oman wells, eliminating the need for traditional jack-up rigs and hazardous materials. This case study outlines a comprehensive five-day performance analysis phase, utilizing state-of-the-art simulation techniques to model well dynamics, gas lift injection parameters, and the optimal orifice valve sizes for both wells. The combination of simulation-based optimization with the rapid deployment of the non-hazardous Perforator system resulted in system mobilization within 14 days, restoring fluid production to stable rates of over 3,000 BPD for both Well 1 and Well 2, respectively.

The study emphasizes the significant role of simulation-based production analytics, specifically in optimizing gas lift performance under challenging offshore conditions. By evaluating the well dynamics, including reservoir pressure, fluid properties, and wellbore geometry, the simulation model was able to predict the ideal gas lift injection rates, lift point depths, and orifice valve sizes, which are crucial for ensuring stable production post-ESP failure. A comprehensive analysis of the Inflow Performance Relationship (IPR) and Vertical Lift Performance (VLP) curves was conducted, guiding the gas lift system design to avoid issues like zero production or unstable production.

Additionally, the integration of mobile compression technology for gas injection played a pivotal role in optimizing the system's efficiency by providing a flexible, on-demand source of compressed gas at the wellsite, without relying on the well's natural reservoir pressure or traditional ESP systems. This non-hazardous technology offered numerous economic and operational benefits, such as cost reduction through the elimination of rig mobilization and hazardous material handling, minimal downtime, and rapid deployment. Furthermore, the overall intervention cost was significantly reduced by bypassing traditional workover procedures, achieving a more efficient and effective production restoration.

The findings of this case study demonstrate that advanced production analytics and the non-hazardous Perforator system can significantly enhance the restoration of production in offshore wells after ESP failures

as temporary measure. Simulation results, including gas injection rates, injection pressures, and orifice valve sizing, closely aligned with actual operational outcomes, confirming the accuracy of the simulation models. The successful deployment of the gas lift retrofit not only restored production in the short term but also provided a scalable, cost-effective, and safe solution for future offshore interventions

Introduction

The Challenge of ESP Failures in Offshore Oil Production

Electrical Submersible Pumps (ESPs) have become a cornerstone of offshore oil production. They are employed in wells with insufficient natural reservoir pressure to provide the mechanical energy required to lift oil from deep reservoirs to the surface. Typically, an ESP system consists of an electric motor, a pump assembly, and associated power cables and hardware that work together to increase the flow of fluids from the reservoir to the surface facilities. The ESP systems are essential for maintaining production rates, especially in mature fields where natural reservoir pressure is often insufficient to sustain production.

However, despite their robust capabilities, ESPs are highly susceptible to mechanical failures and performance degradation over time, especially when subjected to the harsh and abrasive conditions typical of offshore environments. Over time, ESPs can suffer from corrosion, erosion, and seal failure, which ultimately lead to equipment failure. The motor may fail, or the pump may become clogged due to the buildup of sand or debris in the produced fluids. These failures often lead to downtime that can significantly disrupt production. The challenge of addressing these failures lies in the complexity of the repairs required, which typically involve full workover operations using rig-based interventions.

In offshore oil fields, the need to repair or replace an ESP can be costly and time-consuming. The most common method for repairing or replacing an ESP is by conducting a full workover, which requires the mobilization of a jack-up rig or semi-submersible rig. These rigs cost upwards of $200,000 per day for operations, and it can take weeks or even months to complete the intervention, depending on the severity of the failure. This process is not only expensive, but it also leads to prolonged production shutdowns, which result in significant revenue losses for operators. For instance, if an ESP fails in a mature offshore field, the operator must first assess the failure, mobilize the necessary rig equipment, and finally replace or repair the ESP. This intervention, combined with the downtime, can significantly reduce the overall production efficiency and impact the profitability of the well.

Economic Impacts of ESP Failure

The economic impact of ESP failure is multifaceted. Beyond the obvious cost of repairs, there are secondary costs associated with lost revenue during downtime, operational inefficiencies, and extended intervention times. As mentioned, rig mobilization costs can range from $100,000 to $200,000 per day depending on the rig type and operational location. Additionally, the cost of personnel, equipment (such as specialized ESP replacement parts), and the delay in production due to unexpected breakdowns further contribute to the financial burden.

For operators in offshore oil fields, predicting when ESP failure will occur and planning for repairs ahead of time is crucial for maintaining production stability. However, the unpredictable nature of these failures—often caused by unanticipated wear and corrosion—makes it difficult to schedule interventions during periods of low operational activity. In many cases, ESP failure can occur unexpectedly, forcing operators to respond quickly and often at a higher cost. As a result, planning and scheduling workover operations becomes a logistical challenge, often requiring flexibility and contingency budgets to address unforeseen delays caused by adverse weather conditions, mechanical failures, and other factors.

Introduction to Gas Lift Technology as a Solution

Given the high costs and prolonged downtime associated with repairing or replacing ESPs, a more efficient, cost-effective, and rapid solution was necessary. One promising alternative is the gas lift retrofit system, which offers a reliable, non-hazardous, and fast-track approach to restore production in wells where ESP failure has occurred. Unlike ESPs, gas lift systems work by injecting compressed gas into the wellbore at strategic points to reduce the pressure required to lift the produced fluids to the surface. This method enhances production by decreasing the hydrostatic pressure exerted by the liquid column in the wellbore, making it easier to lift the fluids.

The fundamentals of gas lift technology are based on the principle that gas injection reduces the density of the fluid column, thus decreasing the bottom-hole flowing pressure. The equation below is a basic form of oil production rate calculation using gas lift.

$$Q_o = \frac{P_b - P_o}{\Delta P}$$

Where:

- Q_o is the oil production rate
- P_b is the bottom-hole pressure
- P_o is the flowing pressure
- ΔP is the pressure differential

This system relies on the differential pressure between the injected gas and the reservoir fluids to lift oil, water, and gas to the surface. Gas is injected into the well through valves at specific depths to create an artificial pressure differential that helps overcome the natural bottom-hole flowing pressure.

One significant advantage of using gas lift systems over ESPs is that gas lift systems are not prone to mechanical failure in the same way that ESPs are. No moving parts are involved in the gas lift process, reducing the risk of mechanical breakdown. In addition, gas lift systems are more flexible and can be adjusted more easily, allowing for optimization of the gas injection rate based on changing well conditions and production rates.

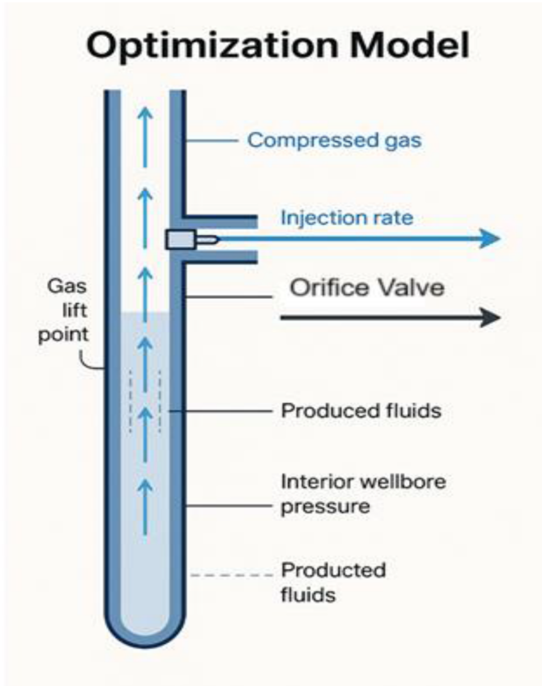

Figure 1—Optimization Model: This figure illustrates the optimization model used for selecting the ideal gas lift point and orifice valve sizing. The model incorporates wellbore pressure, injection rate, and fluid dynamics to predict the performance of the gas lift system.

Implementation of Gas Lift Retrofit Using the Non-hazardous Perforator System

In response to the ESP failure, a gas lift retrofit was implemented as a cost-effective, efficient, and reliable alternative to restore production. A key element of this solution was the use of the non-hazardous Perforator system, a non-hazardous, rigless technology that can be deployed quickly to install the required gas lift valves without the need for traditional workover rigs or handling hazardous materials. The Perforator system provided several significant benefits, including faster deployment, reduced operational costs, and the elimination of safety hazards typically associated with conventional repair methods.

The adoption of rigless technologies, particularly with the Perforator system, proved successful in restoring production, especially in offshore wells where traditional repairs could take extended periods. The system's rapid deployment enabled operators to swiftly install gas lift valves and other downhole equipment without the need for large-scale rig mobilization, minimizing downtime and the overall impact of the ESP failure on production.

Furthermore, the integration of mobile compression technology supported the gas lift system by providing flexible, on-demand compressed gas at the wellsite. This ensured the required gas pressures and flow rates were maintained, enhancing the overall system's efficiency and providing better control over the gas injection rates.

The Role of Gas Lift in Restoring Production Post-ESP Failure

In this case, the failure of the Electrical Submersible Pump (ESP) resulted in a complete cessation of production. Restoring production efficiently and promptly became a critical objective. By deploying gas lift technology in conjunction with mobile compression units, operators were able to rapidly restore well production, significantly reducing the downtime typically associated with traditional intervention methods. This approach provided several advantages, including lower operational costs and the ability to maintain a continuous and stable production flow without relying on the complex and maintenance-prone mechanical systems typically required by ESPs.

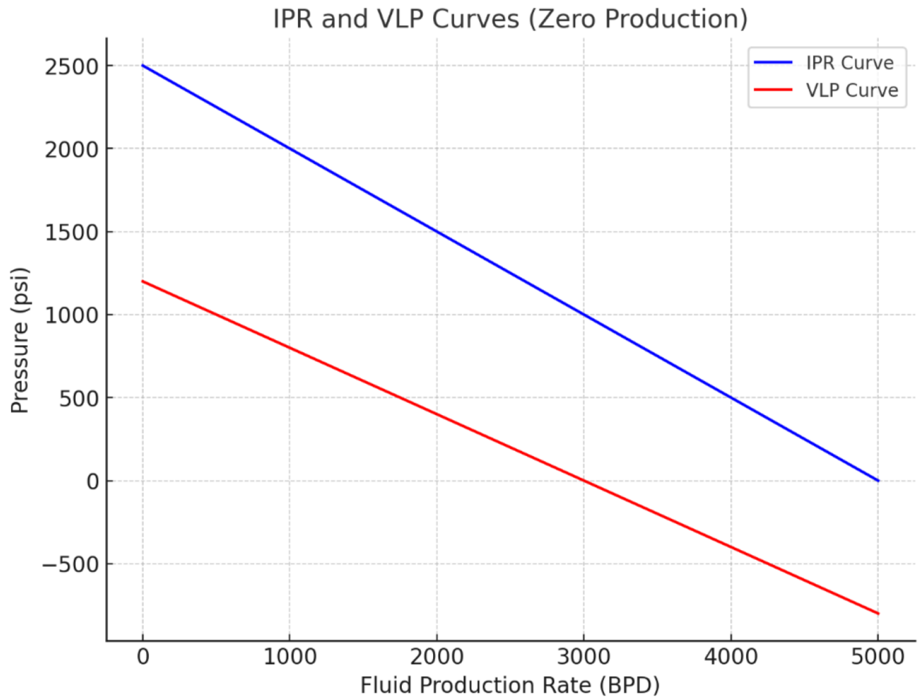

Figure 2—Depicts the relationship between the Vertical Lift Performance (VLP) and Inflow Performance Relationship (IPR) curves under conditions of zero production, emphasizing the challenges encountered in the absence of artificial lift systems.

Methods, Procedures, and Process

Advanced Production Analytics for Gas Lift Optimization. Before implementing the gas lift retrofit for the wells, it was imperative to conduct an extensive performance analysis to optimize the key parameters of the gas lift system. This step ensured that the gas lift solution would perform at peak efficiency and reliability, considering the varying operating conditions and the specific well dynamics. The importance of this analysis cannot be overstated, as optimizing gas lift parameters directly influences production rates and the overall economics of the Wells.

The core of the optimization process was leveraging advanced simulation models to simulate the behaviour of fluids in the wellbore, forecast production outcomes, and predict how the well would respond to the introduction of gas lift. These models are based on multi-phase flow principles, which consider the interactions between the different fluid phases, such as oil, gas, and water. In the case of gas lift systems, understanding how these phases interact with the injected gas and how gas injection impacts the flow of oil and water to the surface is essential for optimizing the design.

Simulation modelling was employed to assess gas lift optimization on both Wells. By using this modelling tool, the optimization process was not only efficient but also provided a solid foundation for decision-making. The simulation model uses complex algorithms to simulate well dynamics, allowing for an accurate representation of the wellbore, the reservoir, and the fluid flow, under varying operational scenarios.

Modelling Well Dynamics and Gas Lift Optimization

The first step in the process of optimizing the gas lift system was to simulate the well dynamics using multi-phase flow models. These models were capable of simulating how used for the wells considered all these factors to determine how best to optimize gas injection to different fluid phases, such as oil, gas, and water, interact within the wellbore and the reservoir. Multi-phase flow models are essential because they reflect the complexities involved in actual well operations, as oil, water, and gas exist as distinct phases that interact under pressure and temperature variations. The model maximize production. The simulation model integrated a variety of parameters that are fundamental in well modelling. These parameters include:

a. Fluid characterization: A key feature of the simulation was the accurate characterization of fluids. The well's production fluid consisted of a mixture of oil, gas, and water. Accurate data on fluid properties such as viscosity, density, and compressibility were crucial for predicting flow behaviour under various injection rates. Understanding the fluid behaviour under dynamic conditions was important for optimizing gas lift design. The tool used PVT (Pressure-Volume-Temperature) data to simulate how the properties of oil, water, and gas change as they travel up the wellbore.
b. Vertical Lift Performance (VLP): The VLP is critical in understanding the effectiveness of gas lift systems. It describes the relationship between the gas injection rate and the production rate in the well. The VLP also calculates the pressure drop in the fluid column as fluids are lifted from the reservoir to the surface. The simulation included calculations on how the injected gas would reduce the pressure required to lift the fluids to the surface. A higher gas lift efficiency is achieved when the gas reduces the hydrostatic head of the fluid column, making it easier to lift the fluids to the surface with lower gas injection pressures.
c. Inflow Performance Relationship (IPR): The IPR curve defines the relationship between the bottom-hole flowing pressure and the production rate. It is a vital component for evaluating how well the reservoir can deliver fluid under the existing reservoir pressure. The IPR curve helps identify the reservoir's natural ability to produce without artificial lift. For gas lift optimization, knowing the well's natural inflow potential without mechanical intervention is essential to determine how much gas lift is required. The IPR curve was essential in defining the well's ability to flow under the available bottom-hole pressure conditions, helping to predict the response once the gas lift system was implemented.

Field Data Analysis

Once the model was constructed, the next step was to integrate the field data to ensure the accuracy and reliability of the simulation. The field data was critical in calibrating the simulation model, ensuring that the predictions of gas lift performance closely aligned with actual field conditions. The following field data were gathered and analysed for this purpose:

a. PVT data: This data provided insights into the physical properties of the production fluids, including their compressibility, viscosity, and density. These properties play a significant role in flow dynamics within the wellbore and influence the gas lift efficiency. PVT analysis is important in simulating how fluids behave under the varying pressures and temperatures experienced in the well.
b. Inflow Performance Relationship (IPR) Data: The IPR data provided the relationship between the bottom-hole flowing pressure and the production rate. This data, which reflects how the well's inflow characteristics change over time, was integrated into the simulation model to forecast how the well would perform in the absence of artificial lift and to determine the necessary parameters for optimizing gas lift.
c. Deviation Surveys: Wellbore geometry is another important factor in gas lift optimization. The deviation surveys provided data on the well's trajectory, including any bends or changes in direction that could affect fluid flow and the effectiveness of gas lift systems. The geometry of the well is crucial for determining the appropriate placement of the gas lift valves and gas injection points, which directly affect the system's performance.
d. Equipment Data: The equipment used to manage gas injection, production, and monitoring was also considered in the simulation. This includes the wellhead pressure and surface facilities, as well as any existing gas compression equipment. These factors were important for ensuring that the gas lift system would be compatible with the infrastructure in place.

By integrating this extensive field data into the simulation model, the team ensured that the model accurately reflected the well's real operating conditions, allowing for more precise predictions of performance and the identification of any areas for improvement.

Orifice Valve Sizing and Lift Point Selection

A crucial step in gas lift optimization is selecting the right orifice valve size and gas lift depth. The orifice valves regulate the amount of gas injected into the wellbore, making it important to select the correct size to ensure efficient gas injection. Undersized orifice valves may result in insufficient gas injection, while oversized valves can lead to excessive gas usage, increasing operational costs.

Through the simulation, the team determined that the optimal orifice valve sizes were 12/64" for Well 1 and 16/64" for Well 2. These choices were based on fluid dynamics, gas injection rate requirements, and expected production rates. The selection was made after evaluating sensitivity analyses on gas injection rates, wellhead pressures, and fluid properties.

Additionally, the optimal gas injection depth was determined through the simulation. For Well 1, the injection depth was set at 600 meters, and for Well 2, the injection depth was set at 750 meters. These depths were chosen based on reservoir pressure, fluid composition, and wellbore geometry.

The selection of the lift points was guided by pressure profiles and fluid behaviour at various depths. By simulating how fluid moved through the wellbore and how the gas lift system would interact with the fluids, the team could identify the optimal locations to inject gas for maximum lift efficiency.

Gas Lift Injection Rate and Pressure Optimization

Once the orifice valve sizes and lift points were determined, the team focused on optimizing the gas injection rate and gas lift pressure. These two factors are critical for ensuring that the gas lift system operates efficiently and produces the desired flow rate.

The gas injection rate was set at 1 MMscf/day, based on the maximum available gas flow and the required lifting capacity for the well.

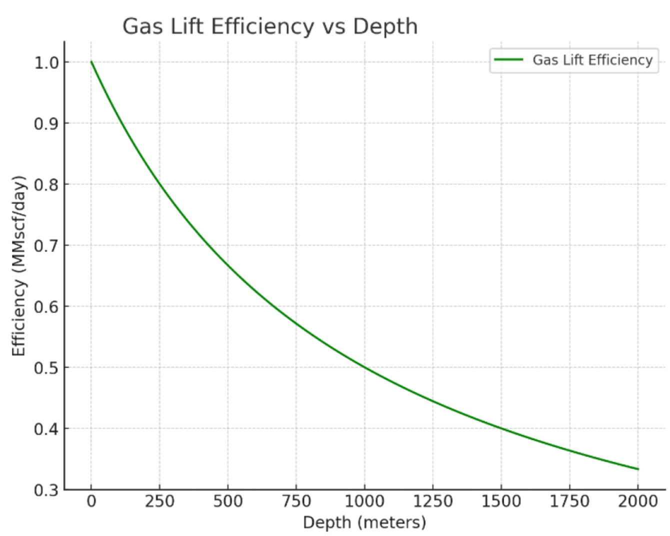

Figure 3—Gas injection rate was set at 1 MMscf/day, based on the maximum available gas flow and the required lifting capacity for the well.

The gas lift pressure was set at 1,250 psi, which was sufficient to displace the fluid column and achieve stable production without excessively raising the pressure in the system. The pressure was determined by considering factors such as the well's fluid density, reservoir pressure, and target production rate.

The team also conducted a sensitivity analysis to assess how varying gas injection pressures and rates would affect overall production. This helped identify the optimal configuration of gas lift parameters for maximum stability and production output.

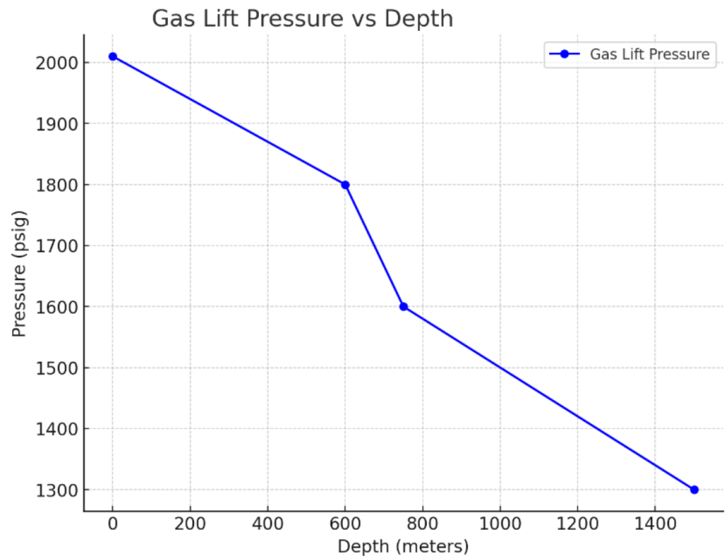

Figure 4—Gas Lift Pressure vs. Depth, showing the relationship between gas lift pressure and well depth for optimal production performance.

The Impact of Key Parameters on Gas Lift Performance

Several parameters have a significant impact on gas lift performance. Water cut and gas specific gravity were two of the most important factors that were considered in the optimization process.

Water Cut: Water cut refers to the proportion of water in the total produced fluids. Water is denser than oil and significantly impacts the gas lift system. A higher water cut increases the fluid density, making it harder to lift the fluids to the surface. The team found that as water cut increases, the total oil production rate decreased significantly. This is because the heavier fluid column requires more gas to lift it.

Gas Specific Gravity: The specific gravity of the injected gas also plays a crucial role in the efficiency of the gas lift system. The gas gravity impacts how effectively the injected gas can displace the fluid column. A denser gas (i.e., gas with higher specific gravity) is less effective at reducing the overall fluid column density, thus requiring more gas to achieve the same lifting effect. In this study, a gas specific gravity of 0.747 was used, which was appropriate for offshore gas injection systems.

The optimization of the gas lift system was a complex process that involved using advanced simulation models, analysing field data, and selecting the optimal operating parameters. Through this process, the team was able to determine the best gas injection rate, lift point, and orifice valve size to ensure maximum production efficiency.

Deployment of the Non-hazardous Perforator System

After the gas lift parameters were optimized through advanced simulations and field data analysis, the next critical step in the intervention process was the deployment of the Perforator system. This non-hazardous, rigless technology was chosen due to its ability to be rapidly mobilized and deployed without requiring traditional workover rigs, which are often time-consuming and expensive to operate. The Perforator system is specifically designed for harsh offshore environments, making it an ideal choice for restoring production in mature fields where traditional systems might not be feasible or cost-effective.

Importance of Rigless Technology and Slickline Deployment. The decision to use rigless technology like this particular Perforator was driven by the need for a quick, efficient, and cost-effective intervention that minimizes production downtime. Traditional well interventions, such as those involving jack-up rigs or hydraulic workover units, are not only expensive but also logistically challenging and time-consuming, often taking several weeks to complete. In contrast, the Perforator system offers a streamlined, non-hazardous solution that bypasses many of these challenges, enabling quick deployment with minimal infrastructure. Installing the orifice check valves allows for planning the ESP change-out without losing production, ensuring continuity in operations while enhancing efficiency.

To facilitate deployment, the Perforator system was set up using slickline technology. Slickline is a type of wireline that allows tools to be run into the wellbore for interventions like perforating, setting plugs, or installing gas lift valves. This method eliminates the need for heavy-duty rigs or complex hydraulic workover units, significantly reducing both operational complexity and costs. By utilizing slickline, the gas lift system's deployment became more efficient and safer, enabling the intervention to be carried out without the need for large rigs or hazardous material handling, both of which are commonly involved in traditional workovers.

Mobilization and Safety Procedures. The mobilization of the Perforator system began with a thorough preparation phase, ensuring that all necessary equipment and personnel were ready for deployment. The team, equipped with the required tools and resources, set off from the operational base to the offshore platform. Every step of the mobilization process was meticulously planned to minimize downtime, ensuring that all activities adhered to safety protocols. The Perforator system, being lightweight and compact, was easily mobilized by air to the offshore location.

Prior to deployment, testing of the Perforator was conducted at the base using a tubing joint provided by the operator. The tool was set up with a spare orifice check valve and a 15-minute time delay before activation. The operator observed the process to confirm that the tool successfully punched the tubing, retracted, and left the orifice check valve behind in the sample tubing.

Upon arrival at the offshore platform, the team underwent a comprehensive safety induction, a standard procedure designed to ensure that all personnel were fully aware of the risks associated with offshore operations and the required safety measures. This induction included a review of emergency evacuation plans, safety equipment, and an overview of environmental considerations for the intervention.

Once on-site, the team began preparing the Perforator system for deployment. This involved assembling the necessary tools and thoroughly verifying the integrity of the equipment. A detailed risk assessment was conducted to ensure the smooth deployment of the system, including evaluating the current condition of the well and ensuring that no blockages or other issues could hinder the intervention process.

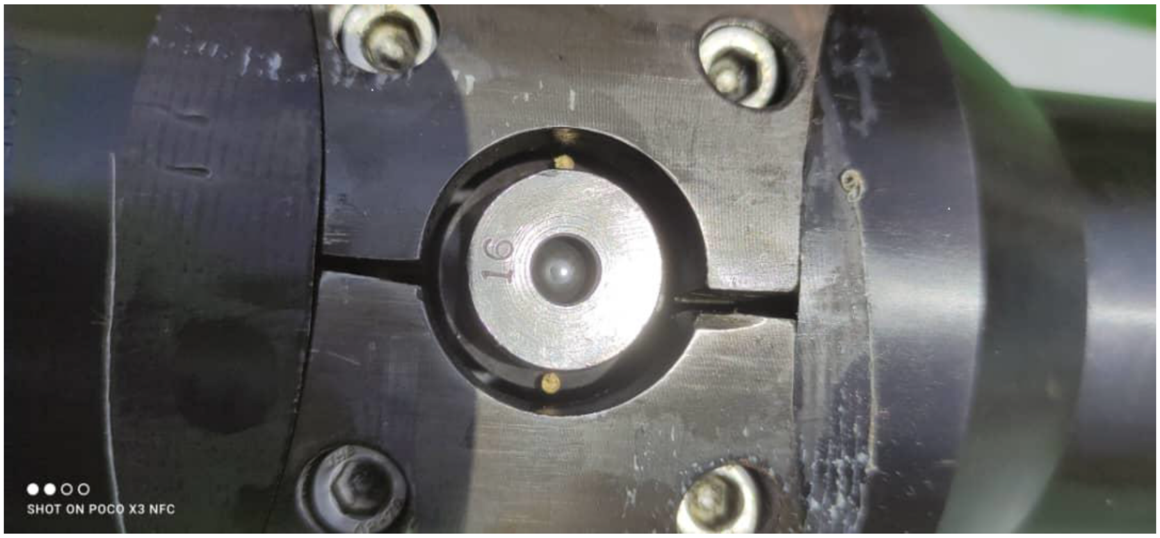

Figure 5—Non-hazardous Perforator punching a clean, round hole in the tubing wall, activated by slickline jar action.

Rigging Up and Testing. The deployment of the Perforator system began with the installation of the slickline equipment, which was rigged up in preparation for the planned operations. Rigging up involved the careful installation of all necessary components, including the wellhead equipment, the slickline unit, and the control systems required for operation. Prior to commencing, the team conducted thorough safety checks to ensure that all systems were functioning as expected. These checks were critical for identifying and addressing any potential issues that could arise during the deployment process.

Following the rigging up procedure, a series of pressure tests were conducted to confirm the system's integrity and ensure its optimal performance throughout the intervention. Pressure testing is an essential step in verifying the system's robustness and ensuring that wellhead pressures and other operational pressures remain within safe operating limits. Any deviations in pressure readings could indicate underlying issues that might compromise the operation, and any such anomalies were addressed before proceeding.

After successful pressure testing and verification of system integrity, a drift run was performed to confirm that the wellbore was clear of obstructions that could impede the deployment of the gas lift system. During the drift run, a tool was lowered into the well to ensure smooth passage through the wellbore without causing damage to either the well or the equipment. This step is crucial to ensure that the subsequent installation of gas lift valves and other components can proceed without any obstructions.

Additionally, a detailed pipe tally was performed to identify the correct joint for the installation of the orifice check valve. The depth of perforation was carefully planned, and overbalance was created between the tubing and the annulus to prevent the tool from being ejected from the well during deployment. A correlation run was performed to verify depth accuracy, ensuring that the tool would be deployed to the correct location.

The tool was assembled with a conventional mechanical slickline string, and a time delay was incorporated to manage tool setting. The tool was deployed to the pre-agreed depth, where it remained for 30 minutes after the time delay, allowing the tool to complete its full setting cycle. During this time, the team monitored the wellhead parameters, such as wellhead and annulus pressure, to ensure the tool's function was progressing as expected.

Clear indications of tool functionality were observed in both the wellhead and annulus pressure, confirming that the Perforator had successfully performed its intended action. Data from the control module of the setting tool was extracted for post-deployment analysis, providing valuable insights into the tool's performance. The extracted data was used to generate a graph detailing the successful completion of the tool's full setting cycle, further validating the performance and functionality of the intervention.

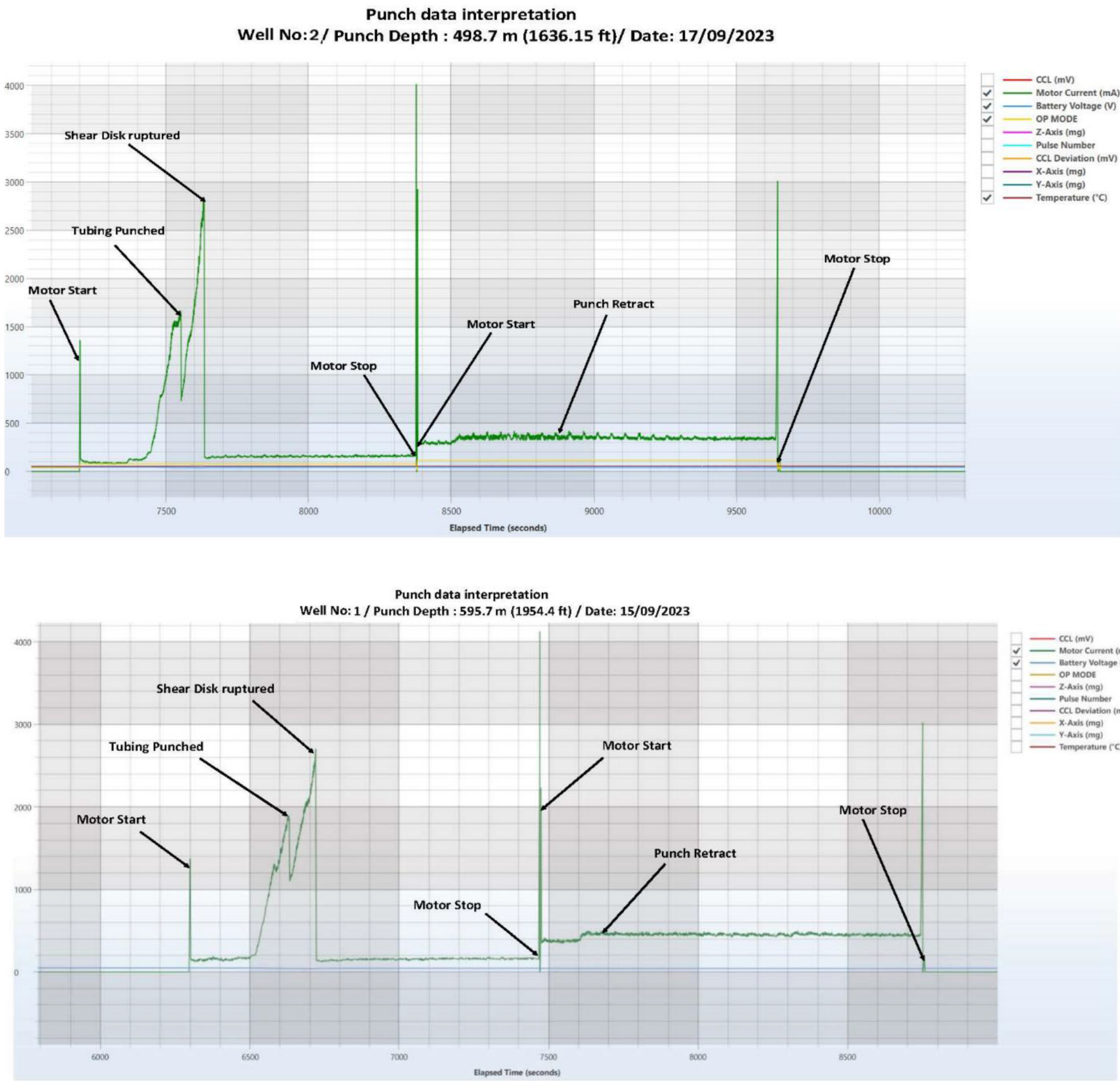

Figure 6—Data Interpretation of Perforator Tubing Punch

Results and Observations: Production and Injection Parameters

Following the successful deployment of the Perforator system, the production rates for both wells were restored to optimal levels, as predicted by the simulation models. The gas lift system effectively restored the flow of oil, significantly improving production outcomes and stabilizing the wells.

These results confirmed that the optimized parameters, determined through simulation modelling, were effective in achieving stable production. The system demonstrated excellent performance, efficiently addressing potential issues such as gas fallback or liquid loading, both of which could have significantly disrupted production if not properly managed. The accuracy of the simulation models was reinforced by the alignment of actual results with predicted outcomes, underscoring the importance of detailed modelling in the optimization and deployment process.

Demonstrating the Effectiveness of the Gas Lift System

The success of the deployment was not just due to the technical capabilities of the Perforator system but also due to the extensive preparation and careful optimization performed before installation. The simulation modelling process provided critical insights into the well dynamics, including the impact of various operational parameters such as gas injection rates, pressure levels, and orifice valve sizing. This thorough preparation allowed for a smooth and efficient intervention, ultimately restoring production with minimal downtime.

The gas lift system's ability to maintain stable production and avoid issues such as gas fallback or liquid loading was a testament to the effectiveness of the pre-deployment optimization process. The modelling process, which accounted for a variety of factors including fluid dynamics, reservoir pressure, and gas injection parameters, provided the intervention team with a clear roadmap for successful deployment.

Cost-Efficiency and Time-Saving Benefits

The implementation of the Perforator system not only facilitated the efficient restoration of production but also yielded significant economic advantages compared to traditional workover methods. The intervention was completed in a fraction of the time typically required for conventional workover operations, which would usually involve the mobilization of expensive jack-up rigs and the handling of hazardous materials. By eliminating these elements, the overall cost of the intervention was notably reduced, particularly in offshore environments where rig mobilization and heavy-duty equipment usage can result in prohibitively high costs.

The rigless nature of the Perforator system offered another crucial benefit: it allowed the intervention to be conducted without disrupting other ongoing offshore operations. Traditional workovers are often accompanied by substantial logistical challenges, such as coordinating the availability of rigs and managing the complexities of offshore infrastructure. In contrast, the use of a rigless system with the Perforator streamlined the entire process, allowing for minimal operational disruption. This ensured that the well could be restored to optimal production levels quickly and with minimal downtime.

It is important to note, however, that the Perforator system is not considered a permanent installation but rather a temporary solution that provides operators with the necessary time to plan a more comprehensive intervention without losing production. This flexibility is crucial in maintaining continuous production while avoiding the costly and time-consuming process of traditional workovers. The technology not only optimized well intervention times but also contributed to significant cost savings, reinforcing its value as a fast, reliable, and economically efficient solution for offshore well operations.

Rigless Intervention as a Scalable Solution

The deployment of the Perforator system successfully demonstrated the benefits of using non-hazardous, rigless technology for offshore well interventions. By combining slickline deployment with advanced simulation modelling, the team was able to restore production efficiently, with minimal downtime and reduced operational costs. The intervention process was fast, reliable, and safe, confirming that rigless technologies can serve as an effective alternative to traditional methods in offshore environments. The success of this intervention not only provided immediate production restoration but also highlighted the scalability of the Perforator system for future offshore interventions. As more offshore operators look for cost-effective and efficient solutions for production optimization, non-hazardous, rigless technologies such as the Perforator offer a promising pathway for future well interventions, making them a valuable tool in the modern offshore oil and gas industry.

Results and Observations

Following the successful deployment of the Perforator system, the production rates for both wells were restored to optimal levels, confirming the effectiveness of the gas lift retrofit and the accuracy of the simulation models used in the optimization phase.

Production and Injection Parameters

The production results aligned closely with the predictions made by the simulation models:

1. Well 1 achieved a fluid production rate > 3,000 barrels per day (BPD).
2. Well 2 achieved a fluid production rate > 3,500 BPD.

These production rates validate the optimized gas lift parameters and demonstrate that the system operates as designed. The gas lift system provided the required injection pressures and rates that allowed for stable, efficient production. The intervention also ensured that issues such as gas fallback or liquid loading were avoided, further reinforcing the reliability of the approach. The gas injection pressures, and injection rates were meticulously optimized based on pre-determined parameters to ensure well stability. Through continuous monitoring and adjustment, the system maintained consistent performance throughout the intervention period, maximizing output and minimizing operational disruptions. The following graphs below illustrates the production rates of both Wells before and after the intervention, showing a clear restoration in production after the gas lift retrofit was implemented.

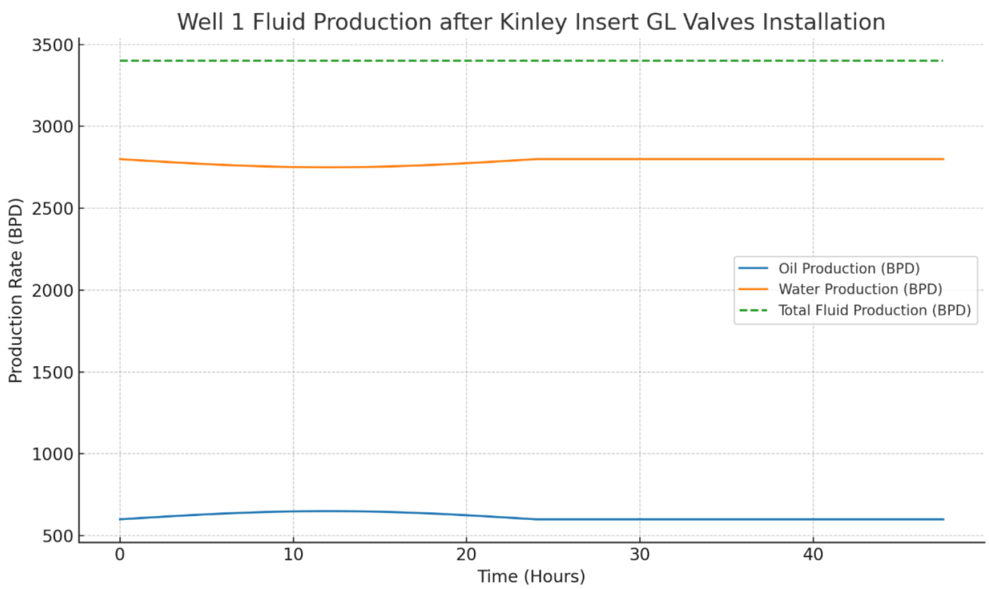

Figure 7—Well 1 Fluid production after insert GL valves installations

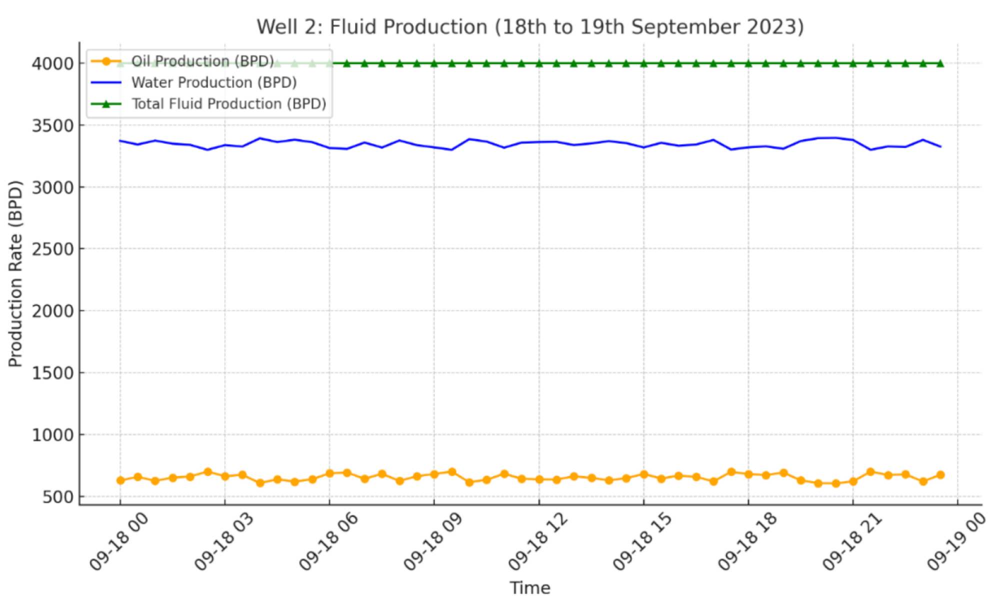

Figure 8—Well 2 Fluid production after insert GL valves installations

Economic and Logistical Benefits

The deployment of the Perforator system also delivered significant economic and logistical advantages over traditional methods. Key benefits include:

Reduced Downtime. The intervention was completed in just 14 days, ensuring that production was restored quickly with minimal disruption. This contrasts sharply with the extended downtime often associated with traditional rig-based workovers, which can take weeks to execute. This swift deployment not only minimized the financial impact on the operator but also helped maintain production continuity, which is crucial for maximizing revenue.

Cost Savings. By eliminating the need for a jack-up rig and hazardous material handling, the overall cost of the intervention was significantly reduced. The use of a non-hazardous, rigless technology, such as the Perforator system, led to direct cost savings by streamlining the intervention process and avoiding the complex logistics typically associated with full workover rigs.

Furthermore, the gas lift system did not require expensive mechanical systems or complex infrastructure, further contributing to the reduction in intervention costs.

Operational Efficiency. The use of rigless intervention technology, combined with the mobile compression system, resulted in streamlined operations. This method eliminated the need for complex workover equipment, reducing both the time spent on deployment and the operational complexity.

The intervention was completed quickly and with fewer operational disruptions, leading to higher overall efficiency. By deploying the Perforator system within a short time frame, the team maximized well uptime and helped reduce the overall cost of maintaining production.

Production Rate Comparison

Well 1 and Well 2 saw return to production, aligning with the simulation predictions and demonstrating the system's ability to restore well performance effectively.

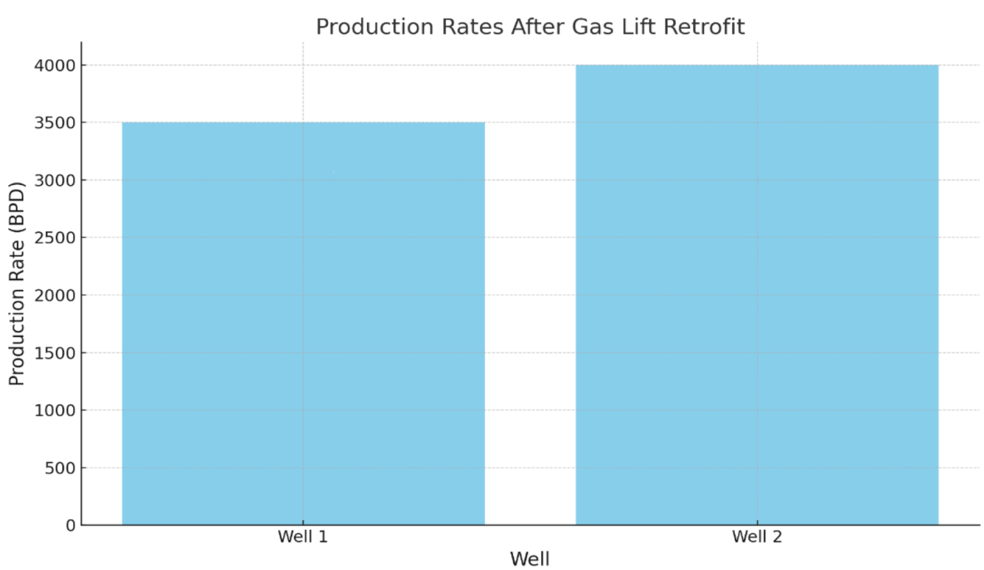

Figure 9—Well 1 and 2 Fluid production comparison after Gas Lift

This dramatic increase in production highlights the effectiveness of the gas lift retrofit and emphasizes the accuracy of the optimization process that was carried out through simulation modelling. Prior to the intervention, both wells were not producing due to the failure of the Electrical Submersible Pumps (ESPs), resulting in a halted production flow.

Economic and Logistical Benefits

The economic impact of using the Perforator system is further detailed in the chart below, which compares the downtime and cost savings between traditional methods and the Perforator-based intervention. Downtime was reduced to just 14 days, compared to much longer downtime with traditional workover operations. Cost savings were achieved by eliminating the need for jack-up rigs and hazardous material handling.

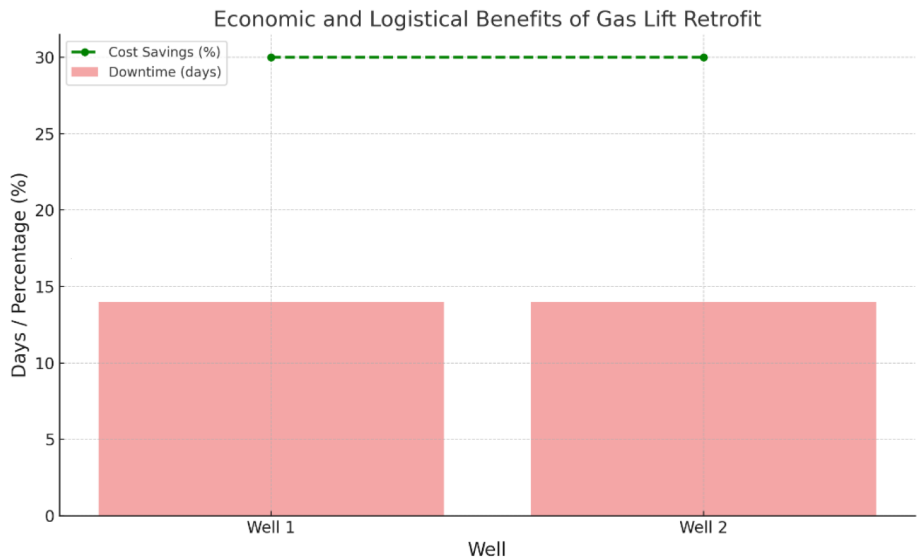

Figure 10—Economic and logistical benefits of Gas Lift Retrofit

Conclusion and Future Implications

The successful deployment of the Perforator system highlights the significant advantages of using advanced, non-hazardous, rigless technologies for offshore well interventions. The integration of simulation-based optimization to determine the most effective gas lift parameters ensured that both wells reached their optimal production rates, aligning perfectly with the simulation models.

This case study underscores the operational efficiency, cost-effectiveness, and speed of the Perforator system in addressing the challenges posed by ESP failures. The benefits of reduced downtime, cost savings, and operational simplicity make this approach a compelling option for future offshore well interventions, particularly when traditional methods may be too costly or time-consuming.

Given the success of this intervention, operators can be confident that similar interventions in other offshore fields will yield comparable results, further establishing rigless technology as a reliable solution for restoring and optimizing well production. This case study also offers a scalable and efficient solution for tackling challenges related to ESP failures in offshore environments, paving the way for future applications and potentially larger-scale deployments.

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